भारी उद्योग मंत्रालय
MINISTRY OF
HEAVY INDUSTRIES
सत्यमेव जयते

SASTRA
ENGINEERING · MANAGEMENT · LAW · SCIENCES · HUMANITIES · EDUCATION
DEEMED TO BE UNIVERSITY
(U/S 3 of the UGC Act,1956)
THINK MERIT | THINK TRANSPARENCY | THINK SASTRA

Electronic Manufacturing System & Product Compliance Training

Ki-CAD PCB Design Assignment

Name :Mahija G

Branch :ECE

SEM/YEAR :3rd/2nd

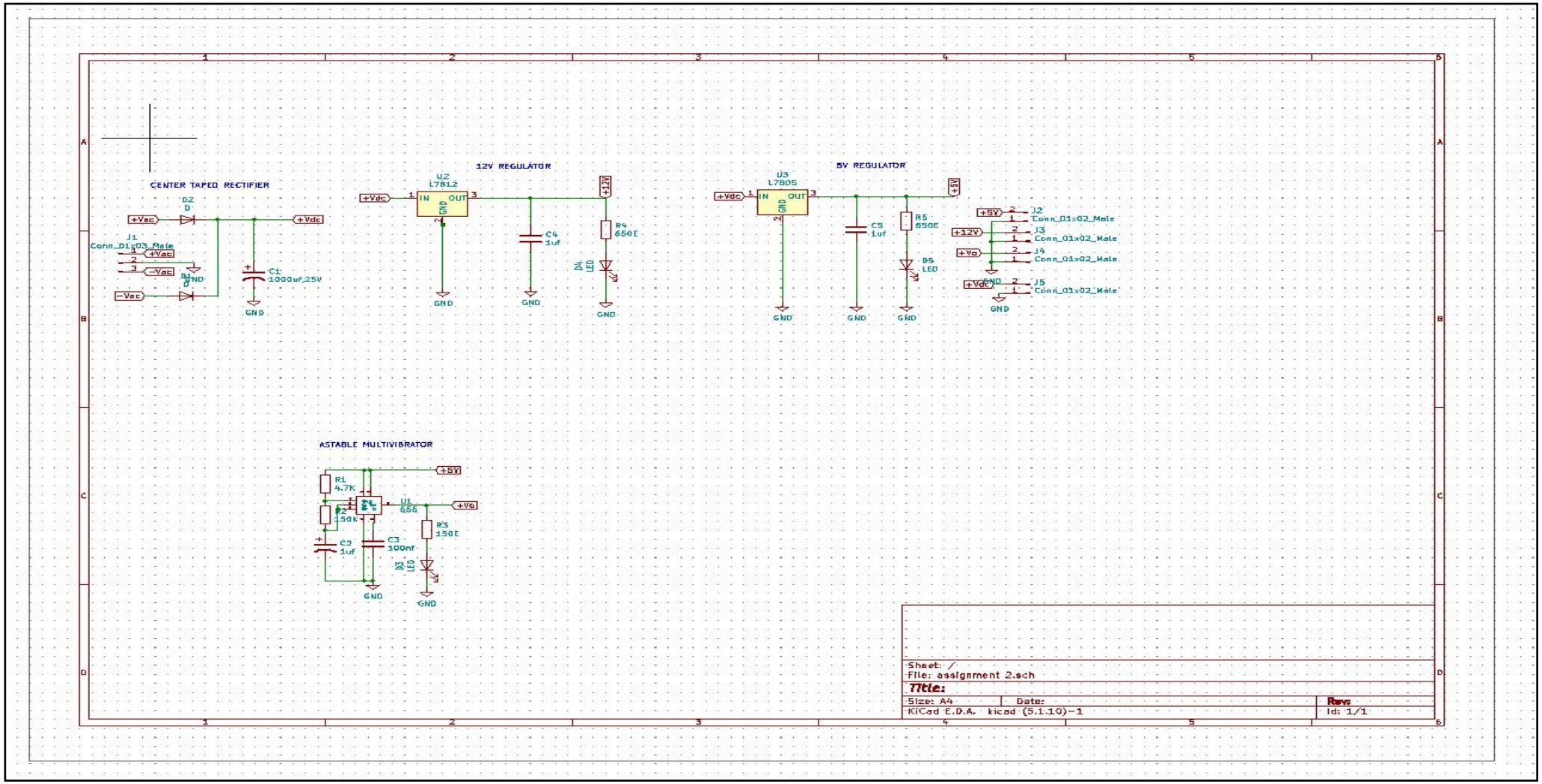
Reg . Number :126004149

Institution Name :SASTRA DEEMED TO BE UNIVERSITY

DESCRIPTION OF THE CIRCUIT

Part	Footprint	Foot Print Image
Diodes	D_0805	
LED	LED_0805	
Resistor	R_0805_2012Metric	
Non Polarized Capacitor	C_0805_2012Metric	
Polarized Capacitor	CP_Elec_8x10.5	
IC 555	SOIC-8_3.9x4.9mm_Pitch1.27mm (Library – Housings_SOIC)	
All connectors	Thro hole	

SCHEMATIC


CENTER TAPED RECTIFIER
12V REGULATOR
5V REGULATOR
ASTABLE MULTIVIBRATOR
U2
L7812
U3
L7805
U1
555
+Vac
-Vac
+Vdc
GND
+5V
+12V
+Vo
C1
1000uf_25V
C2
1uf
C3
100nf
C4
1uf
C5
1uf
R1
4.7K
R2
150K
R3
150E
R4
650E
R5
650E
D2
D
D3
LED
D4
LED
D5
LED
J1
Conn_01x03_Male
J2
Conn_01x02_Male
J3
Conn_01x02_Male
J4
Conn_01x02_Male
J5
Conn_01x02_Male
Sheet: /
File: assignment 2.sch
Title:
Size: A4
Date:
KiCad E.D.A. kicad (5.1.10)-1
Rev:
Id: 1/1

BOM

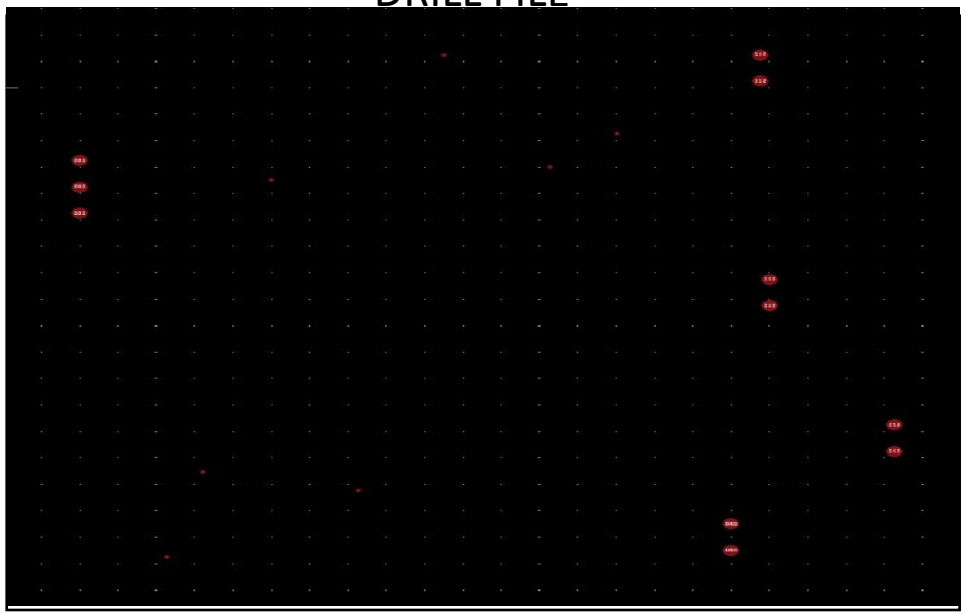
SNo	Component Description	Foot Print	Component Designation	Value	Manufacturer Part Number	Order code of element 14	Unit Price	Reqd qty	Cost for the reqd qty
1		Capacitor_SMD:C_0805_2012Metric	C3	100uF	GRM21BR60G107ME11L	3784457	Rs.32.78	1	Rs.32.78
2		Capacitor_SMD:CP_Elec_8x10.5	c1	1000uF	EEEFT1E102UV	3126815	Rs.159.11	1	Rs.159.11
3		Capacitor_SMD:C_0805_2012Metric	C4,C5	1uF	C0805C105K4RACTU	9227792	Rs.6.20	4	Rs.24.80
4		Package_SO:SOIC-8_3.9x4.9mm_P1.27mm	U1		TLC555ID	3121209	Rs.85.49	1	Rs.85.49
5		Package_TO_SOT_SMD:TO-252-2	U2						
6		Package_TO_SOT_SMD:TO-	U3		L7805ABD2T-TR	1366571	Rs.89.74	1	Rs.89.74

Product Cost

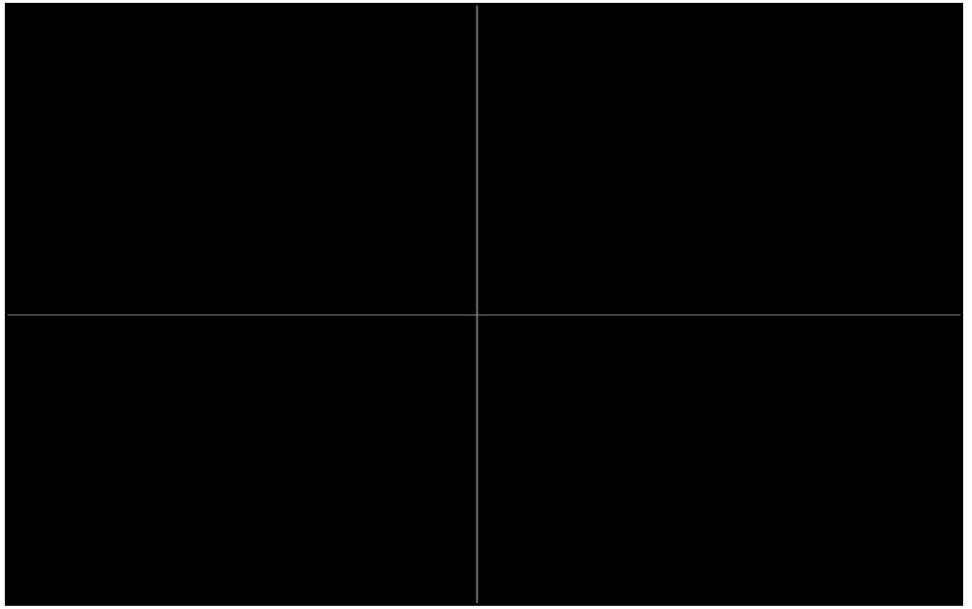
S.No	Description	Cost (Rs)
1	Procurement of Components	700
2	PCB Fabrication Charges	376
3	PCB Assembly Charges	2616
	Total Cost	3692

GERBER FILES

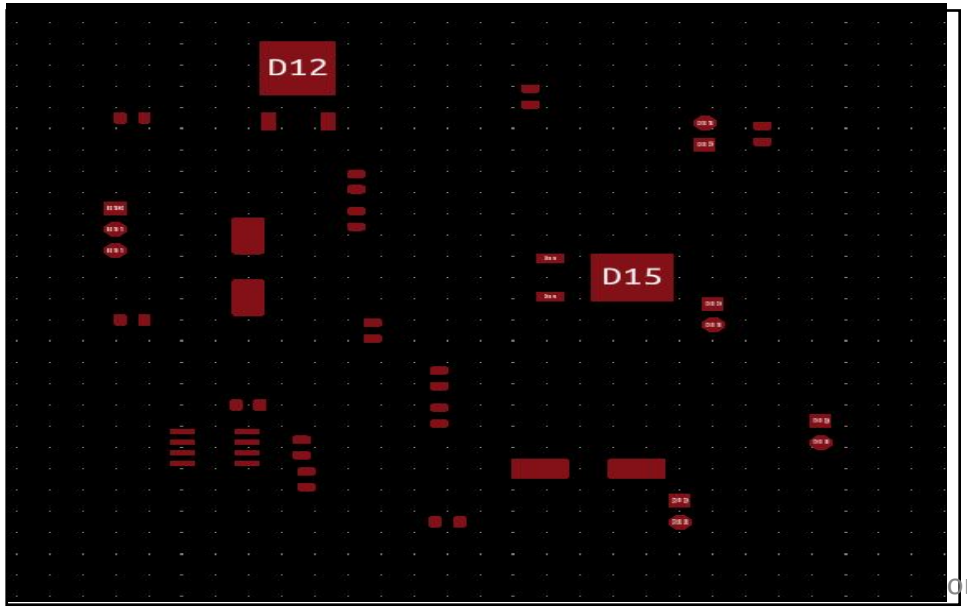
DRILL FILE



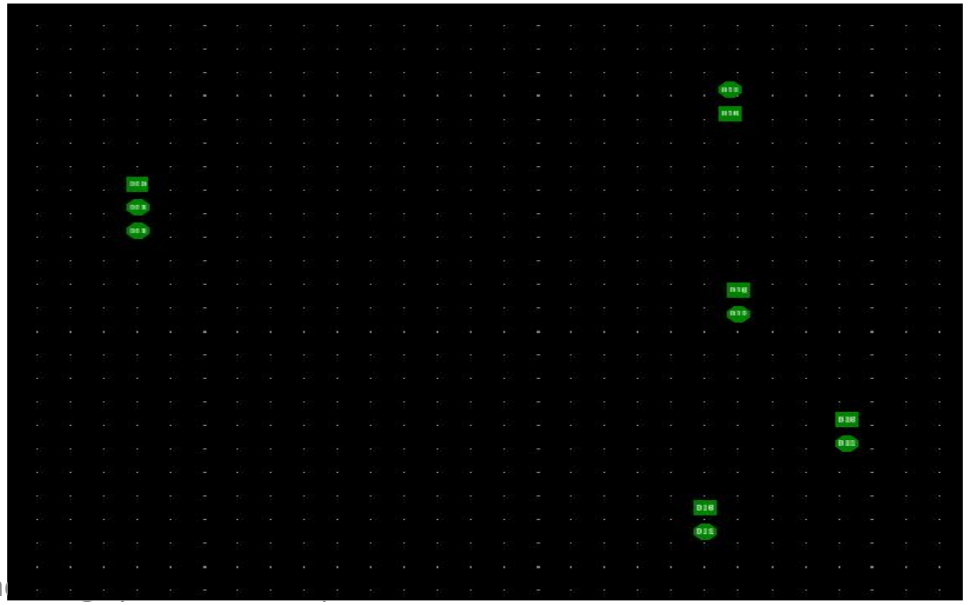
EDGE CUT



FRONT MASK

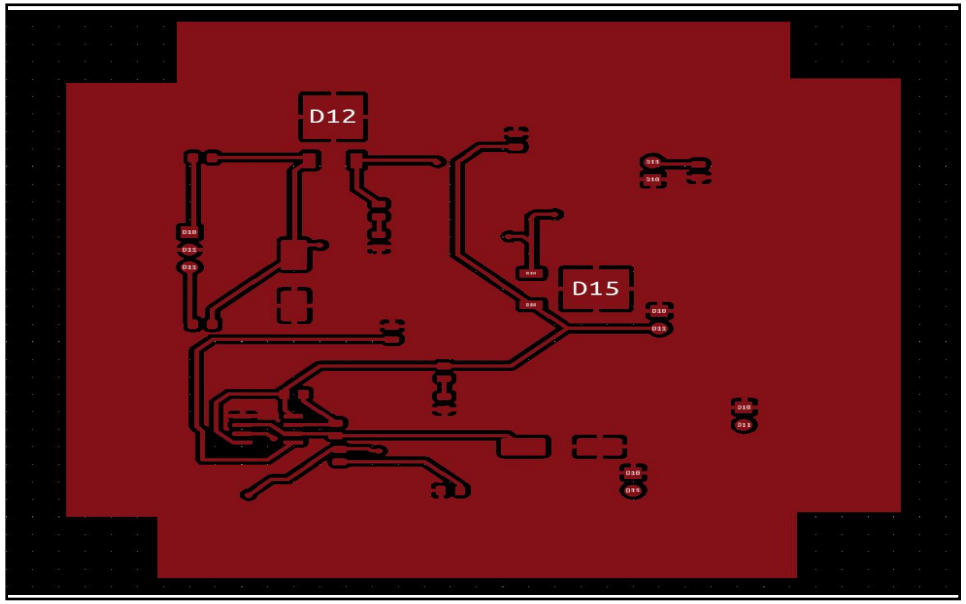


BACK MASK

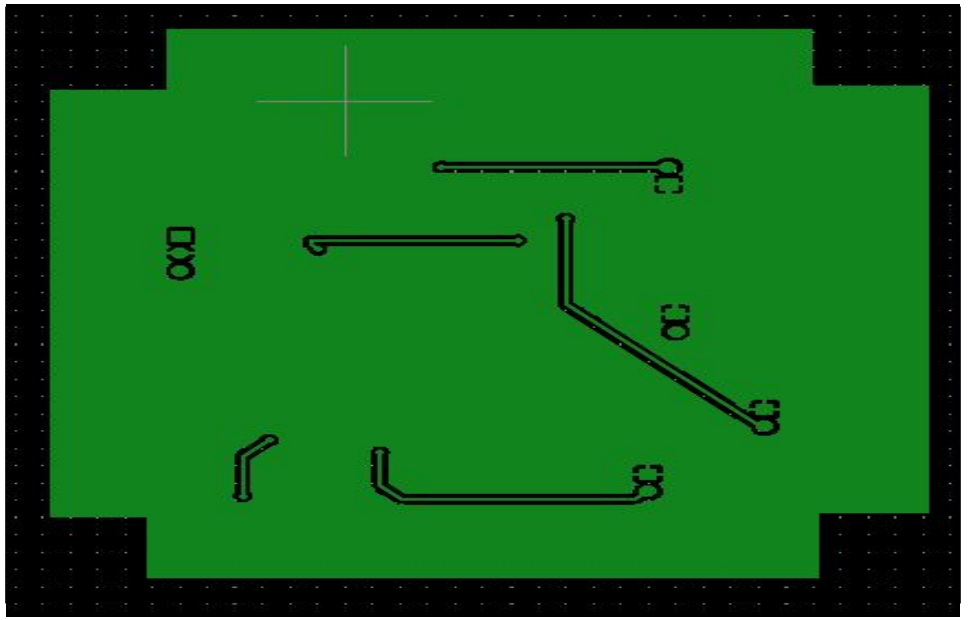


GERBER FILES

LAYER 1



LAYER 2



LAYER 3

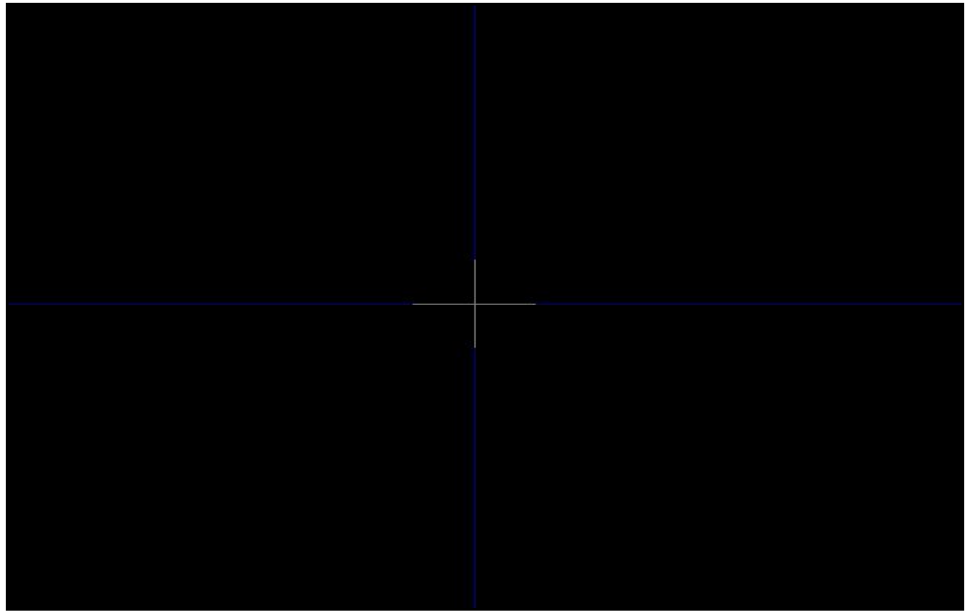
LAYER 4

GERBER FILES

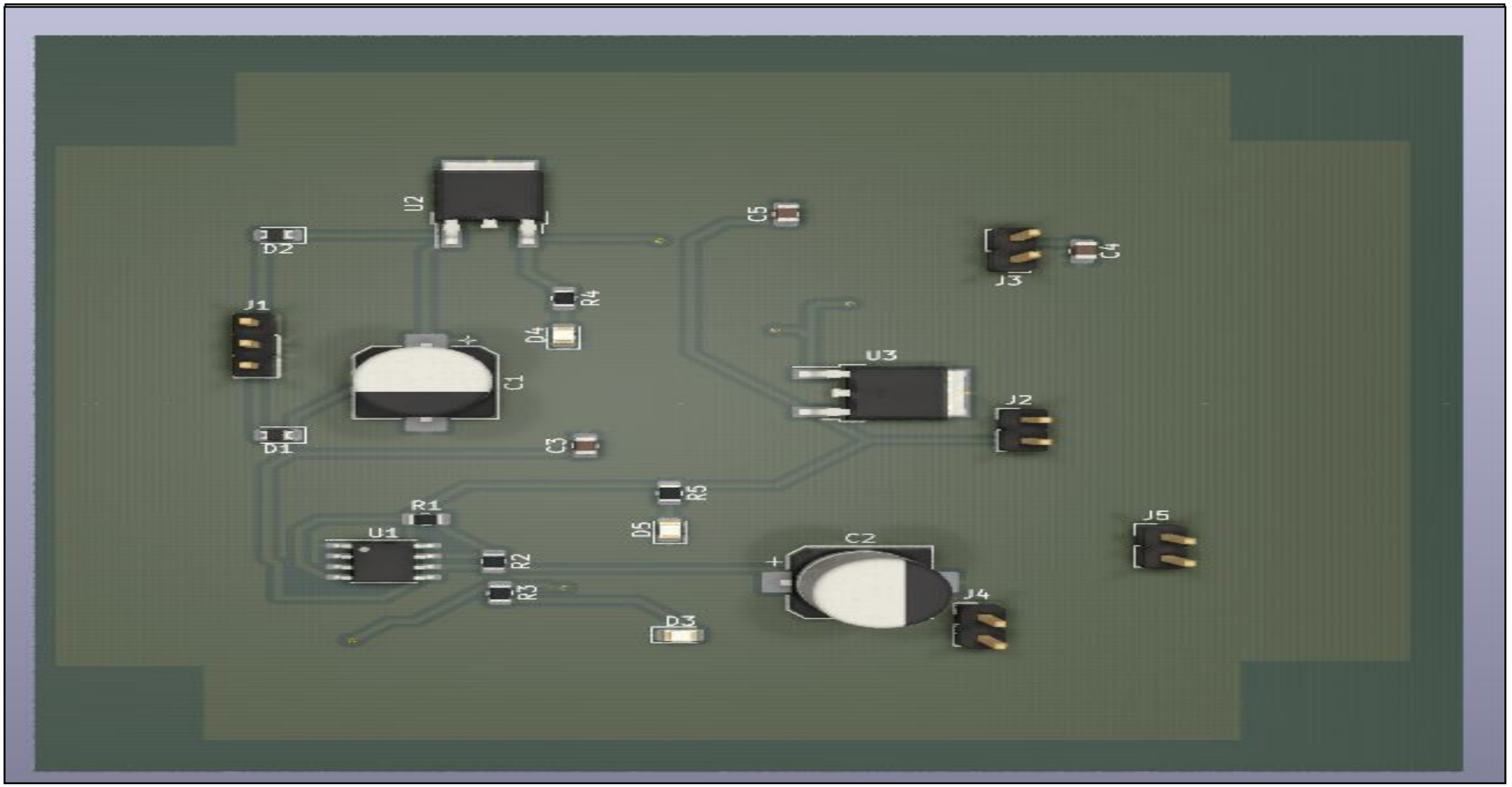
FRONT SILK



BACK SILK



3D VIEW



PCB DESIGN STEPS IN Ki-CAD

Step1: Draw a schematic .
Step2: Assign values to the components used.
Step3: Annotate the schematic.
Step4:Perform electrical rules check.
Step5:Assign footprints to each components.
Step6: Generate netlist.
Step7:Open PCB editor and import the netlist.
Step8:Give routing.
Step9:Generate drill and gerber files.